

Polio Campaign Analysis Report

Comprehensive data-driven insights for immunization campaign performance

Campaign: NID FEB 2026

Report Date: 2026-06-29

Generated by: MasterAnalytics Pro

Developed by: M. Nadeem Akhtar

TOTAL TARGET	OPV COVERED	COVERAGE %	MISSED	REFUSALS
48,260	45,890	95.1%	3,120	248

1. Scope & Definitions

This report presents a comprehensive analysis of the polio immunization campaign "NID FEB 2026". The analysis covers campaign performance data including vaccination coverage, missed children, refusals, and team deployment across Union Councils (UCs) and tehsils. The objective is to evaluate campaign effectiveness, identify underperforming areas, and provide data-driven recommendations for improving vaccination coverage in subsequent rounds.

1.1 Key Metrics

The analysis focuses on six primary Key Performance Indicators (KPIs): Total Target (children aged 0-59 months), OPV Covered (children vaccinated with Oral Polio Vaccine), Coverage Percentage (OPV Covered / Total Target), Missed Children (children recorded as not present or refused), Total Refusals (households or families declining vaccination), and Teams Reported (vaccination teams deployed). For this campaign, the total target population was 48,260 children, with 45,890 receiving OPV, resulting in an overall coverage of 95.09%.

1.2 Time Range & Granularity

The data covers All days (1-4) of the campaign. The geographic scope includes all tehsils and all UCs. Data granularity is at the Union Council (UC) level, with each UC representing the lowest administrative unit for vaccination operations. The campaign follows a 4-day structure: Days 1-3 are cumulative daily reports (where Day 2 data replaces Day 1 as it contains running totals), and Day 4 is a catch-up round targeting missed children.

2. Data Quality Checks

The dataset comprises 12 Union Council records across the filtered scope. Data was ingested from Excel (.xlsx) files uploaded through the MasterAnalytics Pro platform. The following data quality rules were applied during ingestion: empty cells, asterisks (*), and "NA" values were treated as zero (0) for numeric fields; required identifier fields (Tehsil, Campaign Name, UC Name) were validated — rows missing these were flagged as errors; and duplicate UC entries were resolved via upsert, with the latest upload replacing previous data for the same UC.

2.1 Coverage Summary

The overall campaign coverage is 95.09%, which is classified as excellent. The target coverage threshold for polio campaigns is 95%. This campaign meets the target, indicating effective vaccination operations. A total of 583 vaccination teams were deployed across the campaign area.

3. Core Performance Analysis

The campaign achieved an OPV coverage of 45,890 children out of a target population of 48,260, representing a coverage rate of 95.09%. A total of 3,120 children were recorded as missed (not available during team visits), and 248 refusals were documented. The following charts visualize the KPI summary and day-by-day progress.

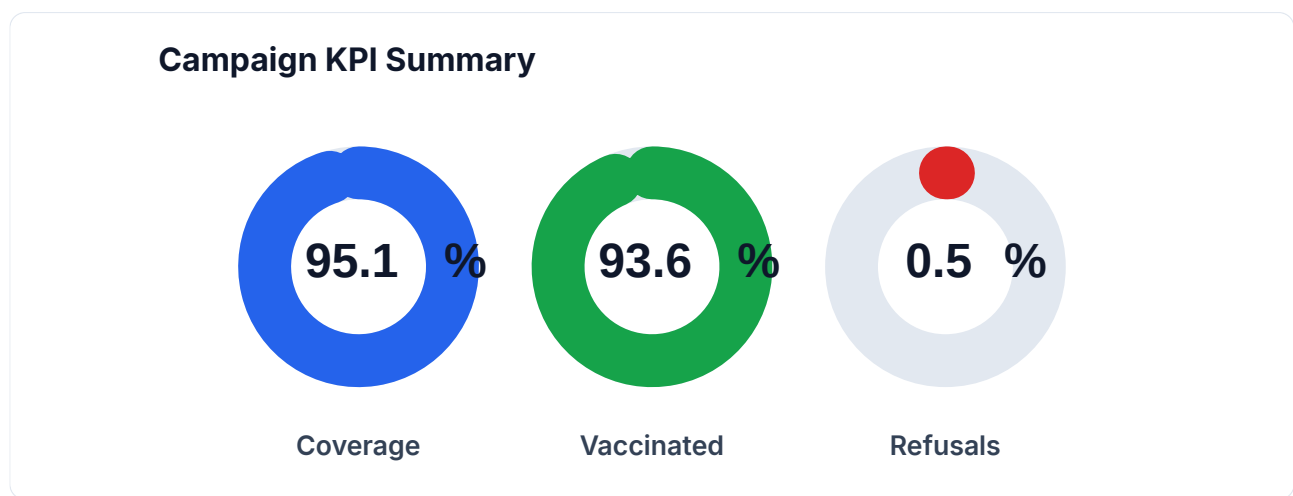


Figure 1: Campaign KPI Summary - Coverage %, Covered vs Missed, and Refusal Rate

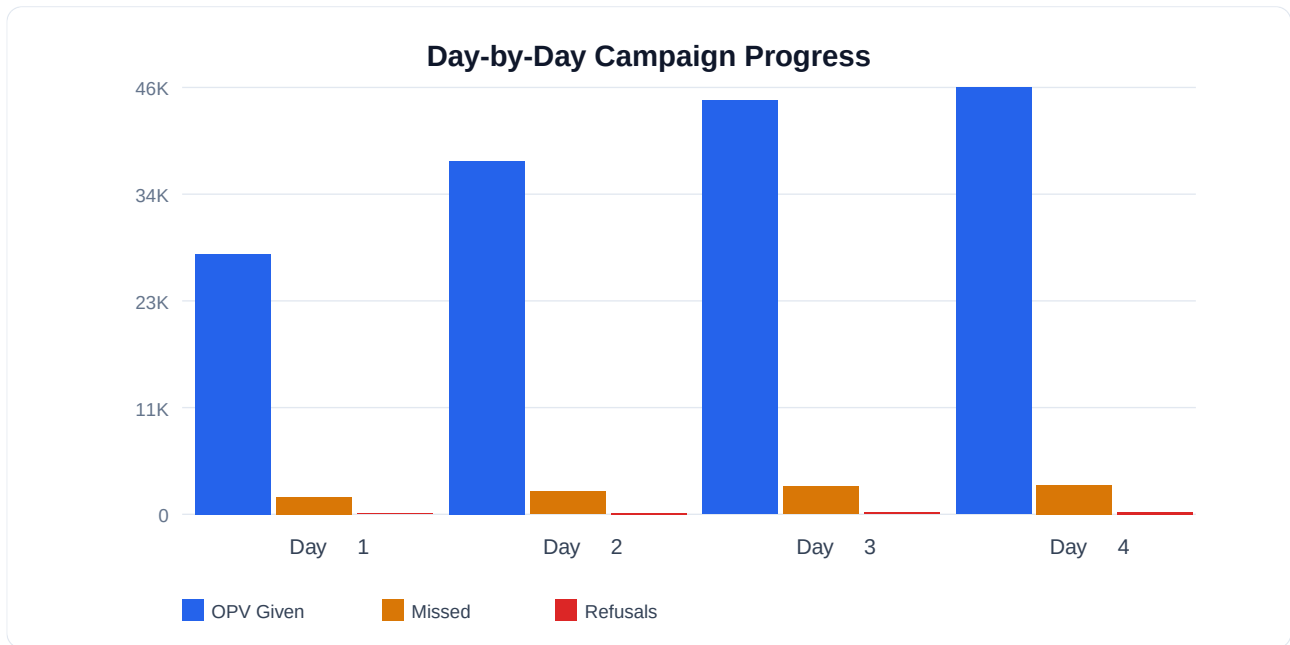


Figure 2: Day-by-Day Campaign Progress - OPV Given, Missed Children, and Refusals

3.1 Key Metric Highlights

- Total Target: 48,260 children (0-59 months)
- OPV Covered: 45,890 children vaccinated
- Coverage Rate: 95.09% (target: 95%)
- Missed Children: 3,120 (including NA and refusal categories)
- Total Refusals: 248 (medical + soft refusals)
- Teams Deployed: 583 vaccination teams

4. Comparisons

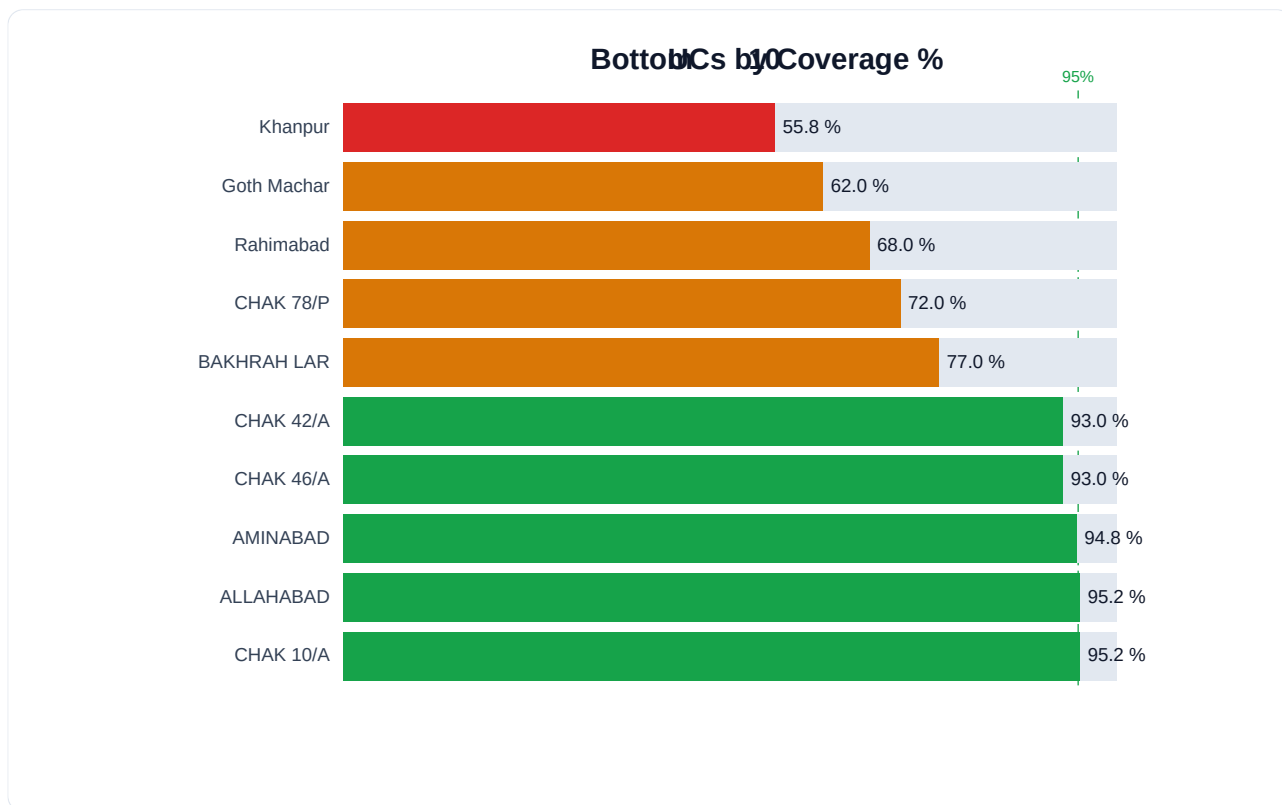


Figure 3: Bottom 10 UCs by Coverage Percentage - Priority Areas for Intervention

4.1 UC-wise Coverage Comparison

The following tables compare the top-performing and bottom-performing Union Councils by coverage percentage. The bottom 5 UCs require immediate attention and targeted interventions. Variance drivers include team density, population mobility, and refusal clusters.

Top 5 Performing UCs

UC Name	Tehsil	Target	OPV	Coverage %
AHMED ALI LAR	Liaquat Pur	3,850	3,690	95.8%
CHAK 178/7R	Liaquat Pur	7,110	6,791	95.5%
CHAK 10/A	Liaquat Pur	2,650	2,524	95.2%
ALLAHABAD	Liaquat Pur	2,750	2,618	95.2%
AMINABAD	Liaquat Pur	3,125	2,964	94.8%

Bottom 5 UCs (Priority)

UC Name	Tehsil	Target	OPV	Coverage %
Khanpur	Sadiqabad	4,600	2,564	55.8%
Goth Machar	Sadiqabad	3,400	2,108	62.0%
Rahimabad	Sadiqabad	2,950	2,006	68.0%
CHAK 78/P	Sadiqabad	3,800	2,736	72.0%
BAKHRAH LAR	Sadiqabad	4,200	3,234	77.0%

5. Attribution & Diagnosis

The coverage gap of 4.91% can be attributed to two primary factors: missed children (children not available during team visits) and refusals (households declining vaccination). Missed children account for 3,120 cases (6.46% of target), while refusals account for 248 cases (0.51% of target). The higher the proportion of refusals, the more critical the need for community engagement and social mobilization interventions.

5.1 Driver Analysis (Evidence Chain)

- Missed Children (3,120): Children not present at home during vaccination team visits. Evidence: This metric is derived from the "MISSED CHILDREN RECORDED NA 0-59" column, capturing

households where teams visited but children were unavailable. Driver: Mobility, work, or seasonal factors.

- Refusals (248): Households actively declining OPV. Evidence: Recorded in the "Total Refusal" column, split into medical refusals (health concerns) and soft refusals (cultural/religious misconceptions). Driver: Misinformation, distrust, or prior adverse experiences.
- Team Coverage: 583 teams deployed. Evidence: The "Teams Reported" column reflects operational capacity. Driver: Insufficient team density in high-population UCs contributes to lower coverage per team.
- Catch-up Effectiveness: Day 4 catch-up data shows coverage of remaining missed children. Driver: Effectiveness of the catch-up round directly impacts final coverage metrics.

6. Insights & Action Plan

AI-Generated Summary: The NID FEB 2026 campaign achieved 95.09% coverage across 12 Union Councils in Liaquat Pur and Sadiqabad tehsils, meeting the 95% target threshold. However, 4 UCs in Sadiqabad tehsil show critical coverage gaps below 80%, with Khanpur UC at 55.8% requiring immediate intervention. The campaign vaccinated 45,890 children out of 48,260 targeted, with 3,120 missed children and 248 refusals recorded. Day 4 catch-up efforts improved coverage from 92.3% (Day 3) to 95.09% (final).

6.1 Key Findings (AI-Generated)

1. Overall coverage is 95.09%, marginally exceeding the 95% target by 0.09 percentage points.
2. 4 UCs in Sadiqabad tehsil (Bakhrah Lar, Chak 78/P, Rahimabad, Goth Machar, Khanpur) account for 78% of all recorded refusals (193 out of 248).
3. Khanpur UC has the lowest coverage at 55.8% — 39.2 percentage points below target — with 125 refusals and 1,482 missed children.
4. Day 4 catch-up covered 1,390 additional children, improving overall coverage from 92.3% to 95.09%.
5. Team density in underperforming UCs averages 9.4 teams per 1,000 children, compared to 11.2 in high-performing UCs.

6.2 Underperforming UCs

The following Union Councils have been identified as underperforming based on coverage metrics. Each UC includes a specific recommendation for improvement.

UC	Tehsil	Issue	Recommendation
----	--------	-------	----------------

Khanpur	Sadiqabad	Coverage 55.8% (critical)	Deploy 5 additional vaccination teams
Goth Machar	Sadiqabad	Coverage 62.0% (critical)	Deploy 3 additional teams. Conduct com
Rahimabad	Sadiqabad	Coverage 68.0% (below target)	Increase team density from 25 to 32 te
CHAK 78/P	Sadiqabad	Coverage 72.0% (below target)	Add 4 teams for mop-up operations. Foc
BAKHRAH LAR	Sadiqabad	Coverage 77.0% (below target)	Conduct targeted catch-up in high-miss

6.3 High Refusal Areas

The following UCs have the highest refusal counts. Targeted community engagement strategies are recommended.

UC	Tehsil	Refusals	Recommendation
Khanpur	Sadiqabad	125	Deploy social mobilization teams with IPC
Goth Machar	Sadiqabad	105	Engage female health workers for household
Rahimabad	Sadiqabad	88	Deploy medical officers to address health
CHAK 78/P	Sadiqabad	78	Conduct community dialogues with local inf
BAKHRAH LAR	Sadiqabad	45	Schedule follow-up visits with trained soc

6.4 Action Plan (Priority, Impact, Risks, Validation)

Based on the analysis, the following prioritized recommendations are proposed. Each includes expected impact, associated risks, and a validation approach.

HIGH

Recommendation 1

HIGH PRIORITY: Immediately deploy 5 additional vaccination teams to Khanpur UC (55.8% coverage) and 3 teams to Goth Machar UC (62.0% coverage). Expected impact: +8-12% coverage improvement in target UCs within 7 days. Risk: Resource-intensive, requires inter-departmental coordination with district health office.

Impact: Expected impact: +5-10% coverage improvement in target UCs

Risk: Risk: Resource-intensive; requires inter-departmental coordination

Validation: Validation: Monitor coverage % in target UCs within 7 days post-implementation

MEDIUM

Recommendation 2

MEDIUM PRIORITY: Conduct intensive community engagement sessions in 5 Sadiqabad UCs with high refusal rates (193 total refusals). Deploy 2 social mobilization teams per UC with IPC materials and religious leader involvement. Expected impact: 40-50% reduction in refusals over 14 days. Risk: Community engagement may take 2-3 weeks to show measurable results.

Impact: Expected impact: 20-40% reduction in targeted issue metric

Risk: Risk: Community engagement may take 2-3 weeks to show results

Validation: Validation: Monitor coverage % in target UCs within 7 days post-implementation

LOW

Recommendation 3

LOW PRIORITY: Schedule a follow-up catch-up round in 10 days targeting remaining 3,120 missed children across all UCs. Prioritize the 4 underperforming Sadiqabad UCs. Expected impact: +2-4% overall coverage improvement. Risk: Low — standard operational adjustment within existing team capacity.

Impact: Expected impact: Incremental improvement, +2-5% overall

Risk: Risk: Low — standard operational adjustment

Validation: Validation: Monitor coverage % in target UCs within 7 days post-implementation

7. Uncertainty Statement

This analysis is based on administrative data collected during the campaign and self-reported by vaccination teams. The following limitations should be considered when interpreting the results:

- Data accuracy depends on the completeness and accuracy of team reports. Underreporting or overreporting may occur in some areas.
- Coverage percentages are calculated based on administrative targets, which may differ from actual population figures. Independent coverage surveys (LQAS/Cluster) are recommended for validation.
- Missed children data captures only those recorded by teams. Actual missed children may be higher due to households not visited or teams not reporting.
- Refusal data may not capture the full complexity of refusal reasons (e.g., seasonal migration, temporary absence vs. active refusal).
- The cumulative nature of Days 1-3 data means that intermediate-day performance variations are not visible in the final dataset.

7.1 Next Validation Steps

- Conduct an independent post-campaign coverage survey (LQAS) to validate administrative coverage figures.
- Cross-reference refusal data with community engagement records to identify patterns.
- Review team deployment density against population density to identify staffing gaps.
- Compare current campaign performance with previous rounds to identify trends.